

## Esercizi termodinamico:

Pag 499 Problemi di riepilogo:

(67)  $L = p/\rho$

$$L_f = 150 \text{ kJ/kg}$$

$$p = 5,5 \text{ MPa} =$$

$$\rho = 1200 \text{ kg/m}^3$$

Percentuale di  
grassi sciolti

$$L = \frac{5,5 \cdot 10^6 \text{ Pa}}{1200 \text{ kg/m}^3} = 4583,3 \text{ J} = 4,583 \text{ kJ}$$

$$Q = L_f m_g \Rightarrow m_g = \frac{Q}{L_f} = \frac{4583,3 \cdot 10^{-3} \text{ kJ}}{150 \text{ kJ/kg}} = 0,0305 \text{ kg}$$

In 1 kg di cioccolato  $m_{\text{grassi}} = 0,3 \text{ kg}$

$$\text{Percentuale} = \frac{0,0305 \text{ kg}}{0,3 \text{ kg}} \cdot 100\% \approx 10,17\%$$

(68) Percentuale = 10% / Q?

$$m_{\text{iceberg}} = 2 \cdot 10^8 \text{ kg}$$

$$L_{\text{acqua}} = 333 \text{ kJ/kg}$$

$$m_{\text{fondere}} = m_{\text{iceberg}} \cdot 0,1 = 2 \cdot 10^8 \text{ kg} \cdot 0,1 = 2 \cdot 10^7 \text{ kg}$$

$$Q = L_f \cdot m = 333 \text{ kJ/kg} \cdot 2 \cdot 10^7 \text{ kg} = 6,66 \cdot 10^{12} \text{ J}$$

(69)  $E_{\text{int, co}} = -1600 \text{ J}$

$$Q_{\text{ob}} = 2000 \text{ J}$$

$L_{\text{obc}}?$

$L_{\text{ob}}?$

$p \uparrow$

$i^c$

$$Q_{bc} = 40 \text{ J}$$

$$L_{ab} = p \Delta V$$

$$L_{bc} = 0$$

$$L_{cba} = L_{ab}$$

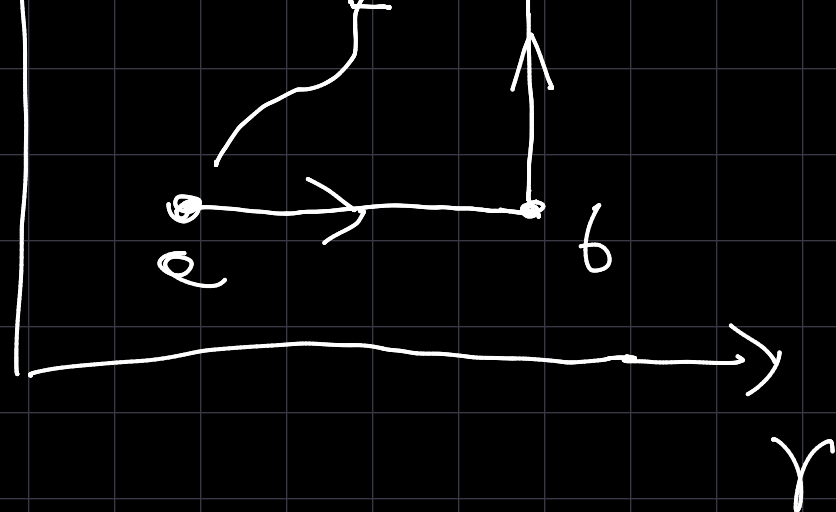
$$E_{\text{ciclo}} = 0 \Rightarrow E_{\text{ciclo}} = E_{ab} + E_{bc} + E_{ca}$$

$$0 = Q_{ab} - p \Delta V + Q_{bc} + E_{ca}$$

$$p \Delta V = Q_{ab} + Q_{bc} + E_{ca}$$

$$= (200 + 40 - 160) \text{ J}$$

$$= 80 \text{ J}$$



$$70) Q = 1 \cdot 10^6 \text{ kcal} = 4184 \cdot 10^6 \text{ J}$$

$$T_{\text{serie}} = 22^\circ \text{ C}$$

$$T_{\text{ecqua}} = 50^\circ \text{ C}$$

$$d = 1 \cdot 10^3 \text{ kg/m}^3$$

$$Q = m c_p \cdot \Delta T =$$

$$4184 \cdot 10^6 \text{ J} = m \cdot \frac{4184 \text{ J}}{\text{kg}^\circ \text{C}} \cdot (50 - 22)^\circ \text{C}$$

$$m = \frac{4184 \cdot 10^6 \text{ J}}{4184 \frac{\text{J}}{\text{kg}} \cdot 28} = 35714.3 \text{ kg}$$

$$V = \frac{m}{d} = \frac{35714.3 \text{ kg}}{1000 \text{ kg/m}^3} = 35.7 \text{ m}^3$$

$$40^{\circ} \text{ kg/m}^3$$

$$(73) \quad l = 5 \text{ cm} = 0,05 \text{ m}$$

$$T_i = 10^{\circ} \text{ C}$$

$$T_f = 60^{\circ} \text{ C}$$

$$\beta = \beta_2 = 3 \cdot 23 \cdot ^{\circ}\text{C}^{-1} = 69^{\circ}\text{C}^{-1}$$

$$V_i = l^3 = (0,05 \text{ m})^3 = 0,000125 \text{ m}^3$$

$$\Delta V = V_i \cdot \beta \cdot \Delta T = (0,000125 \text{ m}^3) (69^{\circ}\text{C}^{-1}) (50^{\circ}\text{C})$$

$$= 0,43125 \text{ m}^3$$

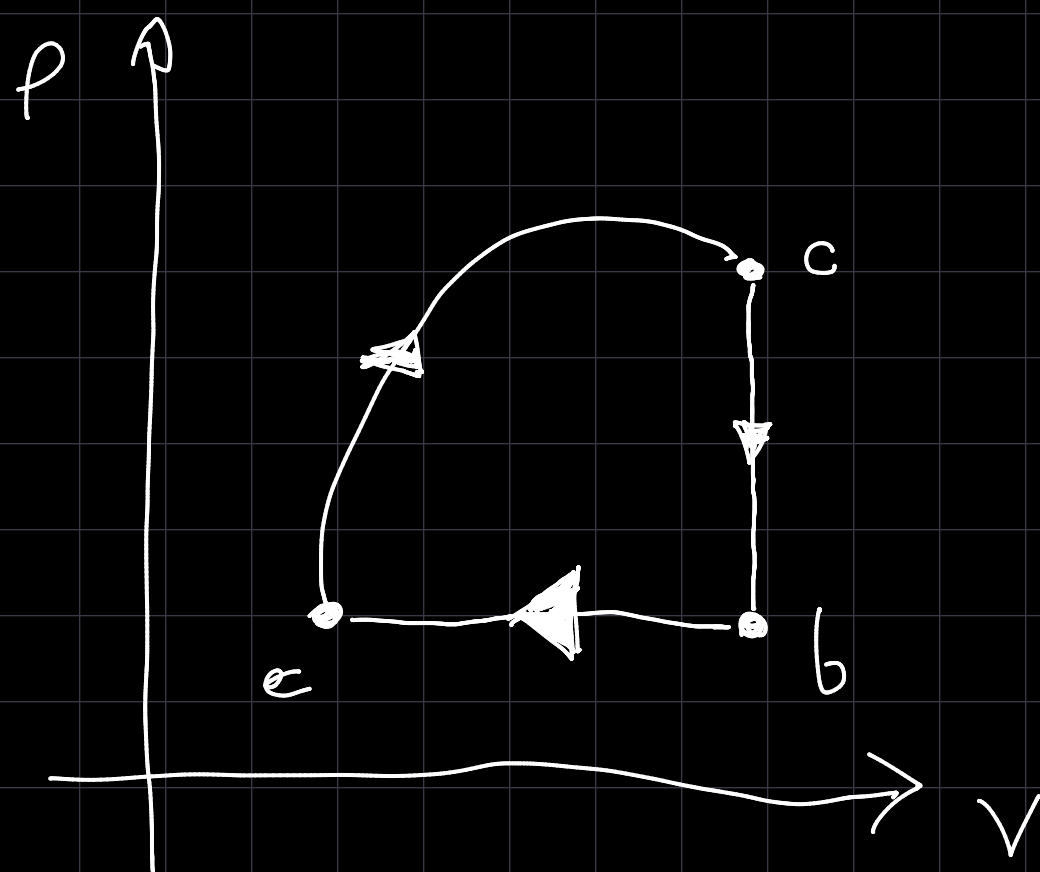
$$(73) \quad Q_{cb} = -40 \text{ J}$$

$$Q_{be} = -130 \text{ J}$$

$$|L_{be}| = 80 \text{ J}$$

$$Q_{ec} = 400 \text{ J}$$

$$L_{ec}?$$



$$L_{cb} = 0$$

$$E_{\text{ciclo}} = 0 \Rightarrow E_{\text{ciclo}} = E_{cb} + E_{be} + E_{ec}$$

$$0 = Q_{cb} - L_{cb} + Q_{be} - L_{be} + Q_{ec} - L_{ec}$$

$$= -40 \text{ J} - 130 \text{ J} + 80 \text{ J} + 400 \text{ J} - L_{ec}$$

$$L_{ec} = 310 \text{ J}$$

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$$++ \quad m = 0,7 \text{ kg}$$

$$T_i = -150^\circ \text{C}$$

$T_f?$

$$Q = 0,6993 \text{ MJ} = 6993 \cdot 10^2 \text{ J}$$

$$c = 2220 \text{ J} / \text{kg} \cdot ^\circ \text{C}$$

$$Q_1 = mc \cdot \Delta T = mc (T_f - T_i)$$

$$= mc (0 - (-150))^\circ \text{C} = 0,7 \cdot \frac{2220 \text{ J}}{\text{kg}^\circ \text{C}} \cdot 150 = 233100 \text{ J}$$

$$Q_2 = L_m = \frac{333 \text{ kJ}}{\text{kg}} \cdot 0,7 = 233,1 \cdot 10^3 \text{ J}$$

$$Q_3 = Q - Q_1 - Q_2 = 6993 \cdot 10^2 \text{ J} - 233100 - 233,1 \cdot 10^3$$

$$= 233100 \text{ J}$$

$$Q_3 = mc \Delta T \Rightarrow \Delta T = \frac{Q_3}{mc} = \frac{233100 \text{ J}}{0,7 \text{ kg} \cdot \frac{4126 \text{ J}}{\text{kg}^\circ \text{C}}} \approx 79,6^\circ \text{C}$$

$$\Rightarrow T_f = 79,6^\circ \text{C}$$

$$(79) \quad p_i = 10 \text{ Pa}$$

$$V_i = 1 \text{ m}^3$$

$$V_f = 2 \text{ m}^3$$

$$p = c V^2$$

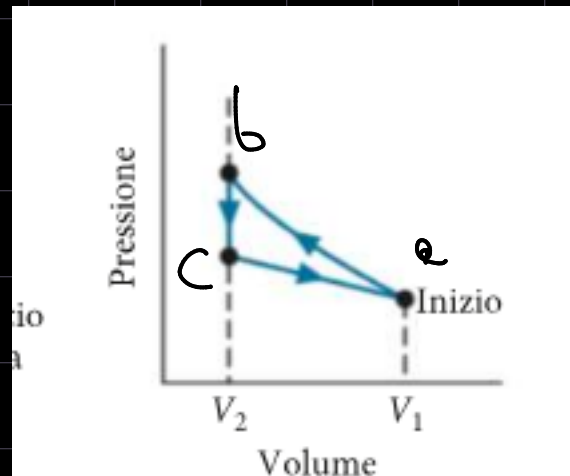
$$c = 10 \text{ N/m}^8$$

$L?$

$$L = p \Delta V = \int_{V_i}^{V_f} p dV = \int_{V_i}^{V_f} c V^2 dV = c \left[ \frac{V^3}{3} \right]_{V_i}^{V_f} =$$

$$= 10 \cdot \frac{8}{3} - 10 \cdot \frac{1}{3} = \frac{80}{3} - \frac{10}{3} = \frac{70}{3} \text{ J} \approx 23,3 \text{ J}$$

80)  $m_{\text{ghiaccio}} = 100 \text{ g} = 0,1 \text{ kg}$  /  $L_{\text{tot}}?$

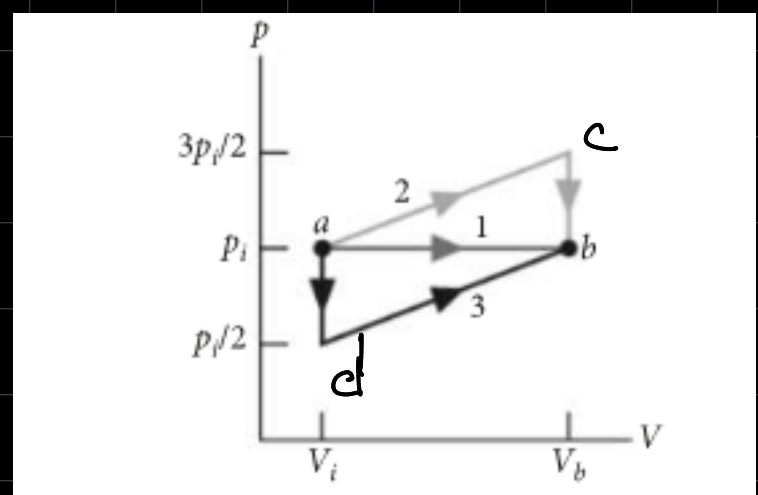


$$Q_{\text{ghiaccio}} = m \cdot L_f = 0,1 \text{ kg} \cdot 334 \text{ kJ/kg} = 33,4 \text{ kJ} = 33400 \text{ J}$$

In un ciclo completo di una trasformazione  $\Delta E_{\text{int}} = 0$

$$\Rightarrow Q - L = 0 \Rightarrow Q = L \Rightarrow L = 33400 \text{ J}$$

81)  $V_b = 5 V_i$  /  $Q_2?$   
 $Q_1 = 10 p_i V_i$  /  $\Delta E_{\text{int}, 3}$   
 $L_1 = p_i \Delta V = 5 p_i V_i - p_i V_i = 4 p_i V_i$



$$\Delta E_{\text{int}, 1} = Q_1 - L_1 = 10 p_i V_i - 4 p_i V_i = 6 p_i V_i$$

$$L_{ec} = \frac{c_1 \cdot c_2}{2} = \frac{ba \cdot cb}{2} = \frac{(3p_i/2 - p_i)(4V_i)}{2} = \frac{1}{2} p_i \cdot 4V_i \cdot \frac{1}{2} = p_i V_i$$

$$L_2 = L_{ec} + L_1 = 5 p_i V_i$$

$$\Delta E_{\text{int}, 1} = Q_2 - L_2 \Rightarrow Q_2 = \Delta E_{\text{int}, 1} + L_2 = 11 p_i V_i$$

$\Delta E_{\text{int}, 3} = \Delta E_{\text{int}, 1}$  in quanto  $\Delta E_{\text{int}}$  non dipende dal percorso ma dal punto iniziale e finale

83)  $T_i = 10^\circ\text{C}$   
 $T_f = 60^\circ\text{C}$   
 $t = 8\text{ cm} = 8 \cdot 10^{-2}\text{ m}$   
 $s = 0,5\text{ cm} = 0,5 \cdot 10^{-2}\text{ m}$

$\Delta V?$

0.064

Disco di Pyrex

$$\beta = 3\alpha = 3 \cdot 3,2 = 9,6^\circ\text{C}^{-1}$$

$$V = \pi r^2 h = 3,14 \cdot (8 \cdot 10^{-2}\text{ m})^2 (0,5 \cdot 10^{-2}\text{ m}) = 0,001\text{ m}^3$$

$$\Delta V = V \cdot \beta \cdot \Delta T = (0,001\text{ m}^3) (9,6^\circ\text{C}^{-1}) (50^\circ\text{C}) \approx 0,48\text{ m}^3$$

85)  $m_{\text{alluminio}} = 2,5\text{ kg}$   
 $T_{\text{el}} = 92^\circ\text{C}$   
 $m_{\text{acqua}} = 8\text{ kg}$   
 $T_{\text{acqua}} = 5^\circ\text{C}$

$T_{\text{eq}}?$

$$C_{\text{el}} = 900\text{ J/kg}^\circ\text{K}$$

$$C_{\text{H}_2\text{O}} = 4187\text{ J/kg}^\circ\text{K}$$

$$Q_{\text{el}} + Q_{\text{H}_2\text{O}} = 0$$

$$m_{\text{el}} C_{\text{el}} T_f - m_{\text{el}} C_{\text{el}} T_i + m_{\text{H}_2\text{O}} C_{\text{H}_2\text{O}} T_f - m_{\text{H}_2\text{O}} C_{\text{H}_2\text{O}} T_i = 0$$

$$T_f (m_{\text{el}} C_{\text{el}} + m_{\text{H}_2\text{O}} C_{\text{H}_2\text{O}}) = m_{\text{el}} C_{\text{el}} T_i + m_{\text{H}_2\text{O}} C_{\text{H}_2\text{O}} T_i$$

$$T_f = \frac{m_{\text{el}} C_{\text{el}} T_i + m_{\text{H}_2\text{O}} C_{\text{H}_2\text{O}} T_i}{m_{\text{el}} C_{\text{el}} + m_{\text{H}_2\text{O}} C_{\text{H}_2\text{O}}} = \frac{37460}{31246} \approx 12^\circ\text{C}$$

86)  $h = 20\text{ cm} = 0,2\text{ m}$

$$b = 30\text{ cm} = 0,3\text{ m}$$

$$T_i = 10^\circ\text{C}$$

$\Delta A?$

$$T_f = 40^\circ\text{C}$$

$$\text{Pannello di vetro} \Rightarrow \alpha = 9^\circ\text{C}^{-1} \Rightarrow \beta = 27 \cdot 10^{-6}^\circ\text{C}^{-1}$$

$$\Delta A = (0,2 \cdot 0,3) \text{ m}^2 \cdot (27 \cdot 10^{-6}) (30^\circ\text{C}) = 486 \cdot 10^{-7} \text{ m}^2$$

$$(90) \quad T_i = 300^\circ\text{K}$$

$$T_f = 3000^\circ\text{K}$$

$$\beta = 3,0 \cdot 10^{-5}^\circ\text{K}^{-1}$$

$\Delta r$ ?

$$\gamma_{\text{termo}} = \frac{4\pi r^3}{3} =$$

$$\Delta V = \gamma \cdot \beta \cdot \Delta T \Rightarrow \frac{4\pi}{3} (r_f^3 - r_i^3) = \frac{4\pi r_i^3}{3} \beta \Delta T$$

$$\frac{r_f^3}{r_i^3} = \beta \Delta T + 1$$

$$r_i^3 = \frac{r_f^3}{\beta \Delta T + 1}$$

$$r_i = \sqrt[3]{\frac{r_f^3}{\beta \Delta T + 1}} = 6 \ 207 \ 723,63 \text{ m}$$

$$\Delta r = r_f - r_i = 163 \ 276,37 \text{ m}$$

$$(95) \quad \gamma_1 = 1 \text{ m}^3$$

$$p_1 = 40 \text{ Pa}$$

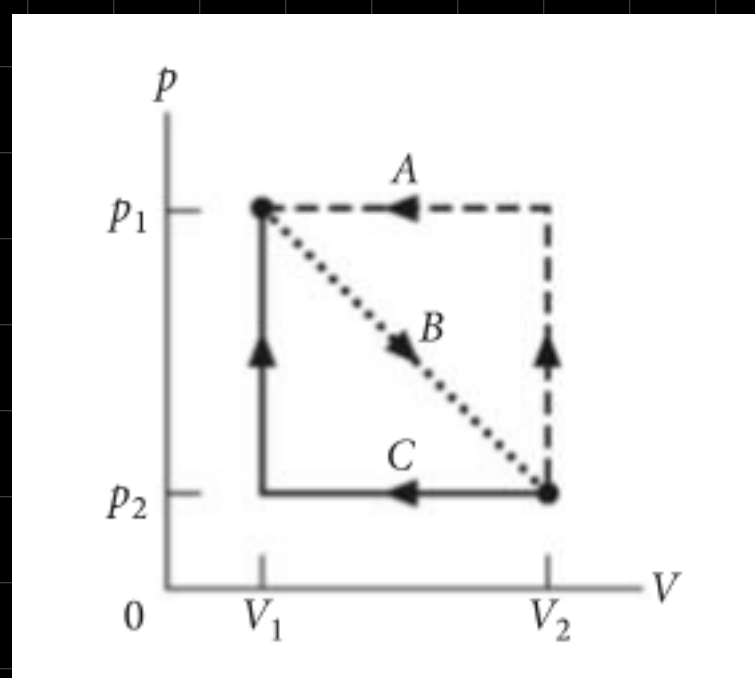
$$\gamma_2 = 4 \text{ m}^3$$

$$p_2 = 10 \text{ Pa}$$

$L_{\text{ciclo}}?$

$L_{BA}?$

$L_{BC}?$



$$L_B = \frac{1}{2} (p_1 + p_2) (\gamma_1 - \gamma_2) = -75 \text{ J}$$

$$L_A = \rho_1 (r_2 - r_1) = 120 \text{ J}$$

$$L_C = \rho_2 (r_2 - r_1) = 30 \text{ J}$$

$$L_{BA} = 45 \text{ J} (L_A + L_C)$$

$$L_{BC} = -45 \text{ J} (L_B + L_C)$$

96)  $L = L_1 + L_2$  /  $\alpha$ ?

con  $\alpha_1$  e  $\alpha_2$

$$\Delta L = \Delta L_1 + \Delta L_2 =$$

$$\Rightarrow L \Delta \alpha = L_1 \alpha_1 \Delta \alpha + L_2 \alpha_2 \Delta \alpha$$

$$\Rightarrow \alpha = \frac{L_1 \alpha_1 + L_2 \alpha_2}{L}$$

b) Se  $L = 52,4 \text{ cm} = 0,524 \text{ m}$

ovvero 1 è acciaio e 2 è ottone

e  $\alpha = 1,3 \cdot 10^{-5} \text{ } ^\circ\text{C}^{-1}$  quanto vale  $L_1$  e  $L_2$ ?

$$\alpha_1 = 11 \cdot 10^{-6} \text{ } ^\circ\text{C}^{-1}$$

$$\alpha_2 = 19 \cdot 10^{-6} \text{ } ^\circ\text{C}^{-1}$$

$$\begin{cases} L \alpha = L_1 \alpha_1 + L_2 \alpha_2 \\ L = L_1 + L_2 \end{cases}$$

$$L = L_1 + L_2$$

$$\begin{cases} L \alpha = (L - L_2) \alpha_1 + L_2 \alpha_2 \\ L_1 = L - L_2 \end{cases}$$

$$L_1 = L - L_2$$

$$\begin{cases} L \alpha = L_1 \alpha_1 + L_2 \alpha_2 \\ \underline{\hspace{2cm}} \end{cases}$$

$$\begin{cases} L (\alpha - \alpha_1) = L_2 (\alpha_2 - \alpha_1) \\ \underline{\hspace{2cm}} \end{cases}$$

$$\left\{ \begin{array}{l} L_2 = L \frac{(a_2 - a_1)}{(a_2 - a_1)} \\ \end{array} \right. ; \left\{ \begin{array}{l} L_2 = \frac{0,524 \cdot 2 \cdot 10^{-6}}{8 \cdot 10^{-6}} = 0,131 \text{ m} \\ L_1 = 0,524 - 0,131 \text{ m} = 0,393 \text{ m} \end{array} \right.$$

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$$m_{\text{ghiaccio}} = 15 \text{ kg}$$

$$T_i = -20^\circ \text{C}$$

$$Q = 7 \cdot 10^6 \text{ J}$$

$$c_g = 2220 \text{ J/(kg} \cdot ^\circ \text{C)}$$

$$L_f = 334000 \text{ J/kg}$$

$$c_{\text{H}_2\text{O}} = 4186 \text{ J/(kg} \cdot ^\circ \text{C)}$$

$$Q_1 = m c_g \Delta T = 15 \text{ kg} \cdot 2220 \text{ J/(kg} \cdot ^\circ \text{C}) (0 - (-20)^\circ \text{C}) = 666000 \text{ J}$$

$$Q_2 = m L_f = 15 \text{ kg} \cdot 334000 = 5010000 \text{ J}$$

$$Q_{\text{rimasto}} = Q - Q_1 - Q_2 = 1324000 \text{ J}$$

$$T_f = \frac{Q_{\text{rimasto}}}{m c_{\text{H}_2\text{O}}} + T_i \approx 21^\circ \text{C}$$

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$$T_i = 0^\circ \text{C}$$

$$\alpha_{\text{acciaio}} = 11 \cdot 10^{-6} \text{ } ^\circ \text{C}^{-1}$$

$$L = 12 \text{ m}$$

$$T_f = 42^\circ \text{C}$$

$$\Delta L = L \cdot \alpha \cdot \Delta T = 225 \cdot 10^{-4} \text{ m}$$

$$d = 2 \cdot \Delta L = 0.00057 \text{ m}$$

Ges perfetti pag 531

64)  $n = 1 \text{ mol}$

$$V_i = 22,4 \text{ L} = 0.224 \text{ m}^3$$

$$V_f = 16,8 \text{ L} = 0.168 \text{ m}^3$$

$$p = 10^5 \text{ Pa}$$

$$L = p \Delta V = (10^5 \text{ Pa}) (0.168 - 0.224 \text{ m}^3) = -5600 \text{ J}$$

65)

$$p_i = 10^5 \text{ Pa}$$

$$V_i = 1 \cdot 10^3 \text{ m}^3$$

$$T = 0,0^\circ \text{C}$$

$$p_f = 10^{10} \text{ Pa}$$

$$V_f = 1 \text{ m}^3$$

$n?$

$T_f?$

$E_{int, i}$

$E_{int, f}$

$$p_i V_i = n R T_i \Rightarrow n = \frac{p_i V_i}{R T_i} = \frac{(10^5 \text{ Pa}) (10^3 \text{ m}^3)}{(8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (273,15^\circ \text{K})} =$$

$$\approx 44055.3 \text{ mol}$$

$$T_f = \frac{p_f V_f}{n R} = \frac{(10^{10} \text{ Pa}) (1 \text{ m}^3)}{(8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (44055.3 \text{ mol})} \approx 27315^\circ \text{K}$$

$$E_{int, i} = \frac{3}{2} n R T_i = \frac{3}{2} (44055.3 \text{ mol}) (8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (273,15^\circ \text{K})$$

$$= 150000135,3 \text{ J}$$

$$E_{int, f} = \frac{3}{2} n R T_f = \frac{3}{2} (44055.3 \text{ mol}) (8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (27315^\circ \text{K})$$

$$= 1.5 \cdot 10^{10} \text{ J}$$

67  $T_i = 330^\circ \text{K}$

$$p_i = 6 \cdot 10^5 \text{ Pa}$$

$$V_i = 500 \text{ cm}^3 = 5 \cdot 10^{-4} \text{ m}^3$$

$$V_f = 1500 \text{ cm}^3 = 15 \cdot 10^{-4} \text{ m}^3$$

Isoterma:  $p_f$ ?  $L$ ?

Adiabatica:  $p_f$ ?  $L$ ?

Risolvo:

Isoterma:  $pV = nRT \Rightarrow n = \frac{p_i V_i}{RT_i} = \frac{(6 \cdot 10^5 \text{ Pa})(5 \cdot 10^{-4} \text{ m}^3)}{(8.31 \text{ J/(mol} \cdot \text{K)})(330^\circ \text{K})}$

$$\Rightarrow n \approx 0.11 \text{ mol}$$

$$p_f = \frac{nRT_f}{V_f} = \frac{(0.11 \text{ mol})(8.31 \frac{\text{J}}{\text{mol} \cdot \text{K}})(330^\circ \text{K})}{(15 \cdot 10^{-4} \text{ m}^3)} = 201102 \text{ Pa}$$

$$L = nRT \ln\left(\frac{V_f}{V_i}\right) = (0.11 \text{ mol})(8.31 \frac{\text{J}}{\text{mol} \cdot \text{K}})(330^\circ \text{K}) \ln(3) = 331.4 \text{ J}$$

Adiabatica:

$$p_i V_i^\gamma = p_f V_f^\gamma$$

$$p_f = p_i \left(\frac{V_i}{V_f}\right)^\gamma = (6 \cdot 10^5 \text{ Pa}) \left(\frac{5 \cdot 10^{-4} \text{ m}^3}{15 \cdot 10^{-4} \text{ m}^3}\right)^{1.67} = 95798.5 \text{ Pa}$$

$$T_f = \frac{p_f V_f}{nR} = \frac{(95798.5 \text{ Pa})(15 \cdot 10^{-4} \text{ m}^3)}{(0.11 \text{ mol})(8.31 \frac{\text{J}}{\text{mol} \cdot \text{K}})} = 157.2^\circ \text{K}$$

$$\Delta E_{\text{int}} = Q - L \Rightarrow Q = 0 \Rightarrow L = -\Delta E_{\text{int}} = -nC_V \Delta T = -n \frac{3}{2} R \Delta T$$

$$= -(0.11 \text{ mol}) \left( \frac{3}{2} \right) (8.31 \text{ J/(mol} \cdot \text{K)}) (177.2^\circ \text{K})$$

$$\approx 2375$$

70

$$n = 2 \text{ mol}$$

$$\Delta T = 15^\circ \text{K}$$

$$\gamma = 1,67$$

trasformazione adiabatica

L?

Q?

$\Delta E_{\text{int}}$ ?

$$L = -n c_v \Delta T = -(2 \text{ mol}) \frac{3}{2} \left( 6,31 \frac{\text{J}}{\text{mol}^\circ \text{K}} \right) (15^\circ \text{K}) \approx -374 \text{ J}$$

Q = 0 in quanto è una trasformazione adiabatica

$$\Delta E_{\text{int}} = -L = 374 \text{ J}$$

73

$$n = 3 \text{ mol}$$

$$c_v = 6 \text{ cal/mol} \cdot \text{K}$$

$$\Delta T = 50^\circ \text{K}$$

Q? L?  $\Delta E_{\text{int}}$

In trasformazione isocora, isobara, adiabatica.

Isocora (V costante):

$$Q = n c_v \Delta T = (3 \text{ mol}) (6 \text{ cal/mol} \cdot \text{K}) (50^\circ \text{K}) = 900 \text{ cal} \\ = 3768,12 \text{ J}$$

L = 0 in quanto V è costante

$$\Delta E_{\text{int}} = Q = 3768,12 \text{ J}$$

Isobara (p costante):

$$Q = n c_p \Delta T = n (c_v + R) \Delta T = (3 \text{ mol}) (25,12 + 6,31) (50^\circ \text{K}) = 5014,5 \text{ J}$$

$$\Delta E_{\text{int}} = n c_v \Delta T = (3 \text{ mol}) (25,12) (50) = 3768,12 \text{ J}$$

$$L = Q - \Delta E_{int} = 5014.5J - 3768.12J = 1246.4J$$

Adiabetica ( $Q = 0$ )

$$Q = 0$$

$$L = -n c_v \Delta T = -3768.12J$$

$$\Delta E_{int} = 3768.12J$$

(76)

$$p = 250 \text{ Pa}$$

$$V_i = 0.8 \text{ m}^3$$

$$V_f = 0.2 \text{ m}^3$$

$$T_i = 360^\circ \text{K}$$

$$Q = 210J$$

$$\Delta E_{int}?$$

$$T_f?$$

Trasformazione isobara.

$$p V_f = n R T_f$$

$$p V_i = n R T_i \Rightarrow n = \frac{p V_i}{R T_i} = \frac{(250 \text{ Pa})(0.8 \text{ m}^3)}{\left(8.31 \frac{\text{J}}{\text{mol} \cdot \text{K}}\right)(360^\circ \text{K})} = 0.067 \text{ mol}$$

$$T_f = \frac{p V_f}{n R} = \frac{(250 \text{ Pa})(0.2 \text{ m}^3)}{\left(8.31 \frac{\text{J}}{\text{mol} \cdot \text{K}}\right)(0.067 \text{ mol})} = 29.8^\circ \text{K}$$

$$L = p \Delta V = (250 \text{ Pa})(0.2 - 0.8) \text{ m}^3 = -150J$$

$$\Delta E_{int} = Q - L = 210J - (-150J) = 360J$$

(77)

$$V_f = 3V_0$$

$$p_0 V_0 = n R T_0$$

$$p_f V_f = n R T_f \Rightarrow p_f 3V_0 = n R T_f$$

$$p_0 V_0 = n R T_0 \Rightarrow p_0 V_0 = n R T_0$$

$$\frac{1}{p_0} = \frac{1}{3 p_0} \cdot \frac{p_0}{p_0} = \frac{1}{3} \frac{1}{p_0} = \frac{1}{3} = 0.33$$

$$b) p_2 = 3^{1/3} p_0$$

$$V_1 = 3 V_0$$

$$V_2 = V_0$$

$$p_1 V_1^\gamma = p_2 V_2^\gamma$$

$$\frac{1}{3} p_0 (3 V_0)^\gamma = 3^{1/3} p_0 (V_0)^\gamma$$

$$\left(\frac{3 V_0}{V_0}\right)^\gamma = 3 \cdot 3^{1/3}$$

$$3^\gamma = 3^{4/3}$$

$$\gamma = 4/3 = 1.33 \text{ polietomico}$$

$$79) V_i = 4 \text{ m}^3$$

$$V_f = 3 \text{ m}^3$$

$$n = 3.5 \text{ mol}$$

$$T = 10^\circ \text{C} = 283^\circ \text{K}$$

L)  
Q)

Compressione isoterma

$$L = n R T \ln\left(\frac{V_f}{V_i}\right) = (3.5 \text{ mol}) \left(8.31 \frac{\text{J}}{\text{mol}^\circ \text{K}}\right) (283^\circ \text{K}) \ln\left(\frac{3}{4}\right)$$

$$\approx -2368 \text{ J}$$

$$\Delta E_{\text{int}} = 0 \text{ in quanto } T \text{ è costante} \Rightarrow L = Q \Rightarrow Q = -2368 \text{ J}$$

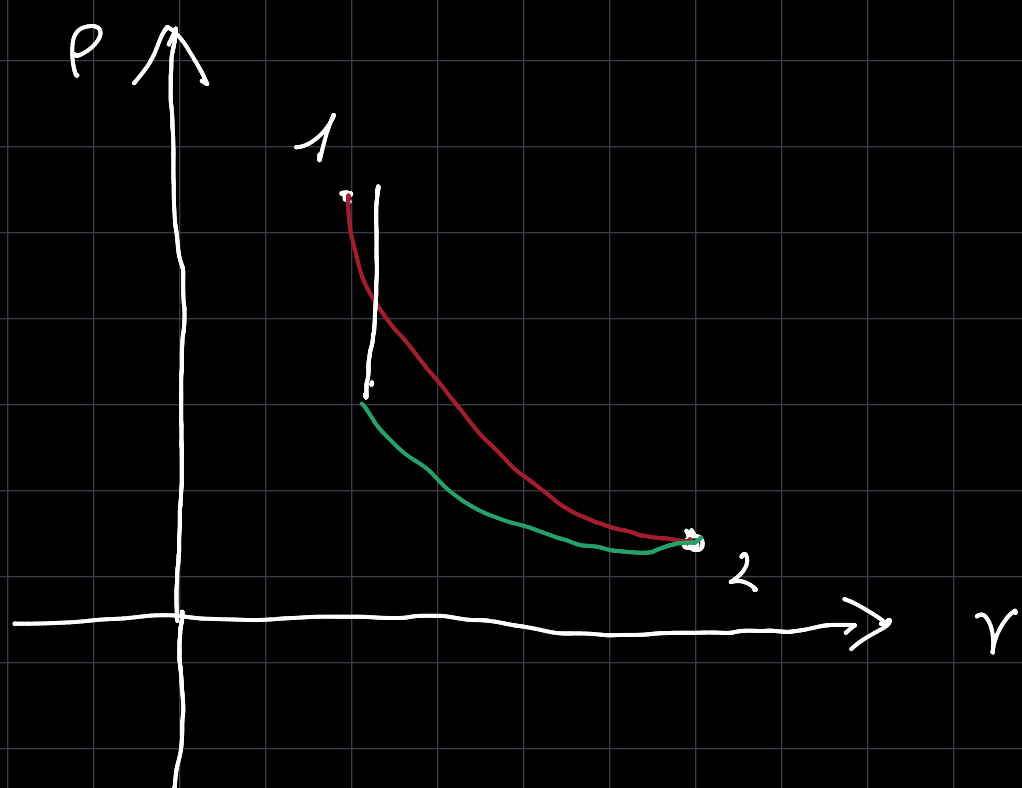
$$81) 1. L_1 = 125 \text{ J (Adiabatica)}$$

$$2. T_2 = 325^\circ \text{K (Isoterma)}$$

$$3. \gamma_{\text{const}}$$

Q<sub>3</sub> ?

Diagramma  $p-v$ :



Nel 1° stadio  $Q_1 = 0$

2° stadio  $Q_2 = n c_v (325 - T_3)$

3° stadio  $Q_3 = n c_v (T_3 - 325)$

$$\Delta E_{int, tot} = 0 \Rightarrow \Delta E_{int, 1} + \Delta E_{int, 2} + \Delta E_{int, 3} = 0$$

$$-125 \text{ J} + n c_v (T_3 - 325) = 0$$

$Q_3 = 125$  e gas (assorbe calore)

83)  $p_i = 32 \cdot 10^5 \text{ Pa}$

$$v_i = 1 \cdot 10^{-3} \text{ m}^3$$

$$v_f = 4 \cdot 10^{-3} \text{ m}^3$$

$$T_i = 300^\circ \text{ K}$$

$p_f$ ?  $T_f$ ?  $L$ ?

Im monoatomico isoterma e adiabatica

Im biatomico e adiabatica?

Monoatomico isoterma:

$$n = \frac{p_i v_i}{R T_i} = 1.28 \text{ mol}$$

$$p_f = \frac{n R T_i}{v_f} = \frac{(1.28 \text{ mol}) \left( 8.31 \frac{\text{J}}{\text{mol}^\circ \text{K}} \right) (300^\circ \text{K})}{4 \cdot 10^{-3} \text{ m}^3} = 797760 \text{ Pa}$$

$$L = n R T \ln \left( \frac{v_f}{v_i} \right) = 4423.7 \text{ J}$$

Monoatomico ed idetico:

$$P_i (V_i)^{1,67} = P_f (V_f)^{1,67} \Rightarrow$$

$$\Rightarrow P_f = P_i \left( \frac{V_i}{V_f} \right)^{1,67} = 316046.5 \text{ Pa}$$

$$L = -n \frac{3}{2} R \frac{P_f V_f}{P_i} + n \frac{3}{2} R T_i \approx +2890.5 \text{ J}$$

Biatomico e diatomico:

$$P_f = P_i \left( \frac{V_i}{V_f} \right)^{1,4} = 459479.3 \text{ Pa}$$

$$L = -n \frac{5}{2} R \frac{P_f V_f}{P_i} + n \frac{5}{2} R T_i = 3382.8 \text{ J}$$

84)

$$n = 3 \text{ mol}$$

$$P_1 = 20 \text{ atm} = 2 \cdot 10^6 \text{ Pa}$$

$$V_1 = 1500 \text{ m}^3$$

$$P_2 = 1,5 P_1$$

$$V_2 = 2 V_1$$

$$P_3 = 2 P_1$$

$$V_3 = 0.5 V_1$$

$$T_1?$$

$$T_2?$$

$$\Delta E_{\text{int}, 1-3}$$

Legge dei gas perfetti:  $PV = nRT$

$$T_1 = \frac{P_1 V_1}{nR} = 120.336.943,4^\circ \text{K}$$

$$T_2 = \frac{P_2 V_2}{nR} = 361.040.830,3^\circ \text{K}$$

$$\bar{T}_3 = \frac{p_3 \bar{V}_3}{n R} = 120336 \text{ g} 43.4^\circ \text{K}$$

$$\Delta E_{\text{int}, 1-3} = n c_v \Delta T = n c_v (\bar{T}_3 - \bar{T}_1) = n c_v (0) = 0 \text{ J}$$

$$(87) \quad p V^\gamma = K \Rightarrow p = \frac{K}{V^\gamma} = p_i \frac{V_i^\gamma}{V^\gamma} = p_i \left( \frac{V_i}{V} \right)^\gamma$$

$$L = \int_{V_i}^{V_f} p dV = \int_{V_i}^{V_f} p_i \left( \frac{V_i}{V} \right)^\gamma dV = p_i V_i^\gamma \int_{V_i}^{V_f} V^{-\gamma} dV = p_i V_i^\gamma \left[ \frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V=V_i}^{V=V_f}$$

$$= p_i V_i^\gamma \frac{V_f^{-\gamma+1}}{-\gamma+1} - p_i V_i^\gamma \frac{V_i^{-\gamma+1}}{-\gamma+1} =$$

$$K = p_i V_i^\gamma = p_f V_f^\gamma \Rightarrow \frac{1}{-\gamma+1} (p_f V_f^\gamma V_f^{-\gamma+1} - p_i V_i^\gamma V_i^{-\gamma+1})$$

$$= \frac{1}{-\gamma+1} (p_f V_f - p_i V_i) = \frac{1}{-\frac{5}{3}+1} (p_f V_f - p_i V_i) = -\frac{3}{2} (p_f V_f - p_i V_i)$$

$$\Delta E_{\text{int}} = n c_v \Delta T = \frac{3}{2} R \frac{p_f V_f}{R} - \frac{3}{2} R \frac{p_i V_i}{R} = \frac{3}{2} (p_f V_f - p_i V_i)$$

$$\Rightarrow L = -\Delta E_{\text{int}}$$